

# Alex McIntosh

Data Scientist & AI/ML Engineer

mcintalmo@gmail.com • www.alexandermcintosh.com • linkedin.com/in/mcintalmo • Minneapolis, MN

## PROFESSIONAL EXPERIENCE

**Data Engineer Associate - Optum**

Aug 2024–Present

- Shipped a multi-agent system using A2A, MCP, and ADK to automate admin tasks
- Automated LLM evaluation with scoring pipelines to monitor consistency, accuracy, safety, latency, and fairness
- Refactored Fabric Spark ETL into reusable patterns, cutting failure rates from 12-15% to <1%
- Mentored an intern team that won first place in an international Shark Tank competition with an agentic AI solution
- Instrumented agents with OpenTelemetry; added tracing and metrics for prompt response, agent calling, and tool use

**Data Science and Engineering Associate - Optum**

Jul 2023–Jul 2024

- Trained a state-of-the-art time-series forecasting model (Prophet + Random Forest) to detect operational anomalies with >90% precision.
- Orchestrated multi-agent AI systems to perform statistical analysis dashboard generation
- Delivered a conversational AI self-service analytics solution
- Enabled observability into millions of updating claims records with Kafka and MLFlow
- Evaluated ML models for accuracy, bias, and operational readiness

**Software Engineer - Constelleum LLC**

Mar 2020–Jul 2023

- Engineered adaptive control schemes enabling inclusive gameplay
- Automated client/server test pipelines
- Analyzed event log data to inform balance and engagement metrics

## EDUCATION & PROFESSIONAL DEVELOPMENT

**Master of Science in Computer Science - Georgia Institute of Technology**

Expected Jul 2027

Machine Learning specialization

**MicroMasters in Statistics and Data Science - MITx**

Dec 2023

**Bachelor of Arts in Physics - Saint John's University**

May 2021

Minored in Music and Mathematics • Phi Beta Kappa Honor Society Member • Men's Chorus President

**Microsoft Certified: Fabric Data Engineer Associate - Microsoft**

Jun 2025

**Microsoft Certified: Azure Fundamentals - Microsoft**

Aug 2024

**Microsoft Certified: Azure Data Scientist Associate - Microsoft**

Jan 2023

## SKILLS & TECHNOLOGIES

Python • PySpark • SQL • R • PyTorch • TensorFlow • Statistics and Probability • Data Science and Predictive Analytics • Natural Language Processing (NLP) • Simulation • Google Vertex AI • MLFlow • Model Context Protocol (MCP) • Agent Development Kit (ADK) • Agent-to-Agent (A2A) Protocol